IB Physics Internal Assessment

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Acceleration in Atwood's Machine

Word Count: 2853

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1 Preliminaries

1.1 Background Theory

Atwood's machine is illustrated in Figure 1.

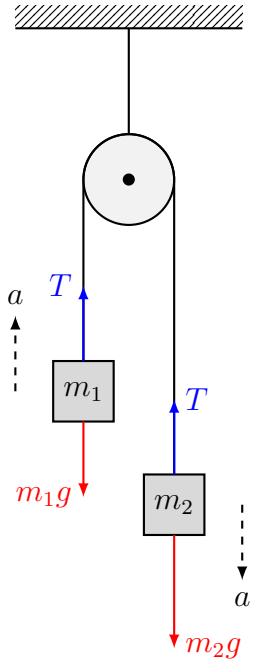


Figure 1: Atwood's machine (self-made).

We assume the pulley is frictionless and massless, the string is massless and inextensible, and tension is uniform throughout the string. We apply Newton's second law, $F_{\text{net}} = ma$, to each mass given $m_2 > m_1$.

For m_2, taking downwards as positive,

$$F_{\text{net}} = m_2 a = m_2 g - T. \tag{1}$$

Solving (1) for T,

$$T = m_2 g - m_2 a. \tag{2}$$

Similarly, for m_1 in Figure 1 [p. 1], taking upwards as positive,

$$F_{\text{net}} = m_1 a = T - m_1 g. \tag{3}$$

Solving (3) for T,

$$T = m_1 g + m_1 a. \tag{4}$$

Given the assumption that tension is uniform across a string, equating (4) and (2) [p. 1] gives

$$m_1 g + m_1 a = m_2 g - m_2 a. \tag{5}$$

Solving (5) for a,

$$m_1 a + m_2 a = m_2 g - m_1 g,$$

$$a(m_1 + m_2) = g(m_2 - m_1),$$

$$a = g\,\frac{m_2 - m_1}{m_1 + m_2}. \tag{6}$$

We denote $(m_2 - m_1)/(m_1 + m_2)$ as the mass ratio M, so

$$a = gM. \tag{7}$$

Comparing (7) with the equation of a straight line, $y = kx + b$ (where $k, b \in \mathbb{R}$), we see that plotting a against M should yield a straight line through the origin with a gradient of g.

1.2 Research question

Thus, our research question reads:

How does the acceleration of an Atwood machine with a plastic pulley and masses m_1 and m_2 vary with the mass ratio $M = (m_2 - m_1)/(m_2 + m_1)$, where $m_2 > m_1$ and $0.06 < M < 0.12$?

We will explain the reasoning behind the range $0.06 < M < 0.12$ in Section 2; briefly, the reason for the lower bound is that frictional effects prevent acceleration below this value. The upper bound was chosen to prevent the masses from accelerating too quickly, which would create a safety hazard. Although the linearisation in (7) [p. 2] does not account for friction, we still specify a plastic pulley, as frictional effects become apparent once our data has been collected and processed. Thus, we control for the material of the pulley to ensure a consistent coefficient of friction between the string and pulley; this is important because the lower bound 0.06 may be too low for a pulley with a different coefficient of friction, which would prevent the system from accelerating at all.

1.3 Hypothesis

Based on the background theory in Section 1.1 [p. 1], we hypothesise that the acceleration of an Atwood machine with a plastic pulley is directly proportional to the mass ratio $M = (m_2 - m_1)/(m_2 + m_1)$, where $m_2 > m_1$ and $0.06 < M < 0.12$.

2 Variables

The dependent variable is the acceleration of the system, a, which has units of $\mathrm{m\,s^{-2}}$.

The independent variable is the mass ratio $M = (m_2 - m_1)/(m_2 + m_1)$, where $m_2 > m_1$. This ratio is dimensionless, since

$$[M] = \frac{[m_2] - [m_1]}{[m_2] + [m_1]} = \frac{\mathrm{kg}}{\mathrm{kg}} = 1.$$

We vary M by changing the values of m_1 and m_2.

Recall (6) [p. 2]:

$$a = g\,\frac{m_2 - m_1}{m_1 + m_2}. \tag{6}$$

If M is large, the system accelerates quickly, creating a safety hazard. This is because one mass accelerates upwards, hitting the pulley. If the acceleration is large, the exerted force might cause the masses to detach and fall. We prevent this by taping the masses together and by providing an upper limit on M.

If M is small, the system accelerates slowly, which decreases the accuracy of the experiment. This is because we let the system go from rest manually, meaning we likely apply a non-negligible force to the system. If the acceleration is already low, the accidentally applied force represents a significant fraction of the net force. This problem is less significant for large M, as the percentage will be smaller.

Thus, we recommend

$$0.06 < M < 0.12$$

to ensure the acceleration of the system is not too high or too low. This range was experimentally determined by trial and error.

2.1 Control Variables

We did not include friction in our model; however, frictional force depends on the coefficient of friction between two surfaces, which depends on the materials that constitute the two surfaces. Thus, the same materials must be used for the pulley (just use the same pulley) and the string throughout the experiment.

3 Method

3.1 Equipment

- Steel rod, $\approx 1.0\,\text{m}$ long and $\approx 1.5\,\text{cm}$ in diameter.
- Clamp to securely hold the rod vertically.
- Hook to attach the pulley to the rod.

- One large pulley or two pulleys combined (as shown in Figure 2).



(a) A single pulley.

(b) Two pulleys combined.

Figure 2: The pulley arrangements used in the experiment.

It is vital that the two pulleys are combined as shown in Figure 2b. This effectively acts as one larger pulley. A single pulley is also sufficient if its diameter is large enough to prevent the masses from hitting each other as one accelerates upwards and the other downwards, which would compromise accuracy. The single pulley shown in Figure 2a was too small for our experiment.

- String, $\approx 3.0\,\text{m}$ long. Such a length is likely excessive; nevertheless, it is useful because the experimenter can cut the string as appropriate.
- Scissors.
- 10+ magnetic masses that are identical (in our case ten $102\,\text{g}$ masses and ten $51\,\text{g}$ masses).
- Scale with precision $\pm\ 1\,\text{g}$.
- Tape.
- Motion sensor.
- Data-logging software.
- Pencil.

3.2 Procedure

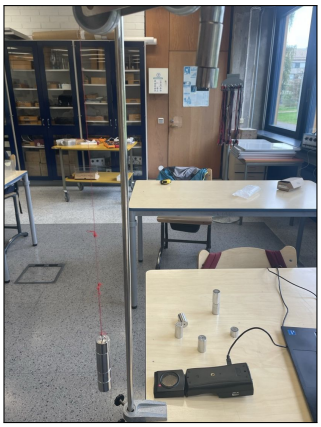
Photographs of our experimental setup are shown in Figure 3.



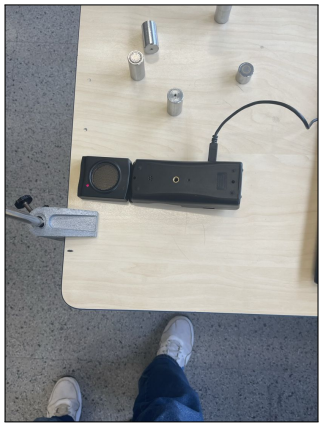
(a) The steel rod held vertically in the clamp, secured to the edge of the lab bench.



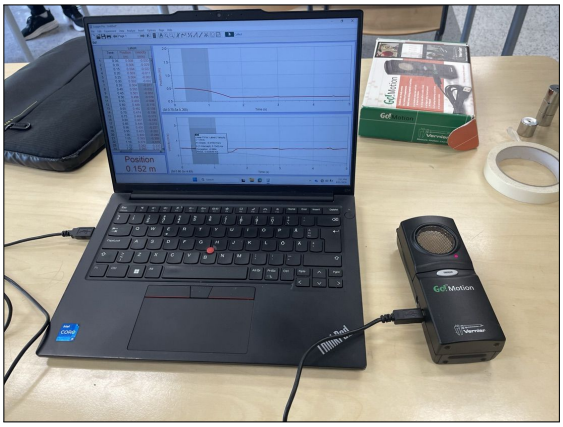
(b) Magnetic masses taped together and hung from the string over the pulley.



(c) The assembled machine, with the motion sensor on the bench below the falling mass.



(d) The motion sensor, positioned directly beneath the masses.



(e) The motion sensor connected to the data-logging software, showing a recorded position trace.

Figure 3: Photographs of the experimental setup.

The procedure is as follows:

1. Gather magnetic masses and measure their masses using the scale. Attach tape to each mass and write its mass on the tape with a pencil.
2. Attach a clamp to a secure surface, such as a table or lab bench (see Figure 3a).
3. Insert the steel rod into the clamp and tighten it. Ensure the rod is high enough for the experimenter to reach the top.

4. Estimate a string length such that, even with many masses at the end, the system cannot get closer than 6 cm to the table. Then cut the string.

5. Attach a hook to the rod and hang a pulley (or multiple attached pulleys as in Figure 2 [p. 5]).

6. Choose a mass ratio as follows.

   Example calculation: We wish to express m_2 as a function of m_1 for a fixed mass ratio M, so that a target combination of masses can be selected graphically. The experimenter can glide through (m_1, m_2) combinations on Desmos or Geogebra to find a suitable combination of masses that is achievable with the available magnetic masses.

   Recall the definition of the mass ratio introduced after (6) [p. 2],

$$M = \frac{m_2 - m_1}{m_2 + m_1}.$$

   Solving for m_2,

$$M(m_2 + m_1) = m_2 - m_1,$$

$$Mm_2 + Mm_1 = m_2 - m_1,$$

$$Mm_2 - m_2 = -m_1 - Mm_1,$$

$$m_2(M - 1) = -m_1(1 + M),$$

$$m_2 = \frac{-m_1(1 + M)}{M - 1}.$$

   Multiplying the numerator and denominator by -1,

$$m_2 = m_1 \left(\frac{1 + M}{1 - M} \right). \tag{8}$$

   Equation (8) is a straight line through the origin with gradient $(1 + M)/(1 - M)$, relating m_2 to m_1 for fixed M.

Worked example: Suppose we choose a target mass ratio of $M = 0.08$. Substituting into (8) [p. 7],

$$m_2 = m_1 \left(\frac{1 + 0.08}{1 - 0.08} \right) = m_1 \left(\frac{1.08}{0.92} \right).$$

Evaluating the gradient,

$$\frac{1.08}{0.92} \approx 1.2.$$

Note: The loss of precision is not significant, as we only need a mass combination close to the selected M. Thus, we have the function

$$m_2 = 1.2\, m_1.$$

Suppose the available magnetic masses are 51 g and 102 g. We search along this line for the nearest achievable combination. Testing $m_1 = 306\,\text{g}$ (three 102 g masses),

$$m_2 = 1.2 \times 306\,\text{g} \approx 370\,\text{g}.$$

The nearest achievable value of m_2 using the available masses is $m_2 = 357\,\text{g}$ (three 102 g masses and one 51 g mass). We select this point, $(m_1, m_2) = (306\,\text{g}, 357\,\text{g})$, and calculate the actual mass ratio achieved:

$$M = \frac{m_2 - m_1}{m_2 + m_1} = \frac{357 - 306}{357 + 306} \approx 0.0769.$$

This value is sufficiently close to the target value of 0.08; thus, we can use $(m_1, m_2) = (306\,\text{g}, 357\,\text{g})$ for our experiment. A value should be rejected if it is outside the range $0.06 < M < 0.12$ or the same M value has already been used in a previous trial.

7. Tape the masses together to prevent the magnetic masses from potentially detaching and falling.

8. Place the motion sensor on the table or lab bench directly below where the masses will fall, and connect it to the data-logging software (see Figure 3d [p. 6] and Figure 3e [p. 6]). The string length was chosen such that the masses cannot approach within 6 cm of the table, ensuring the motion sensor is not damaged.

9. Attach the string to the pulley and hang the magnetic masses from the string (see Figure 3b [p. 6] and Figure 3c [p. 6]).

10. Start the data-logging software and release the mass by hand, attempting to apply as little force as possible to the system.

11. If the sensor did not record the acceleration correctly (this is possible if the masses do not fall directly above the sensor and end up swaying), repeat step 10 until a reading that shows approximately linear velocity (i.e., constant acceleration) is obtained. Then select the section with linear velocity in the data-logging software, calculate the average acceleration for the boxed region shown in Figure 4, and record it.

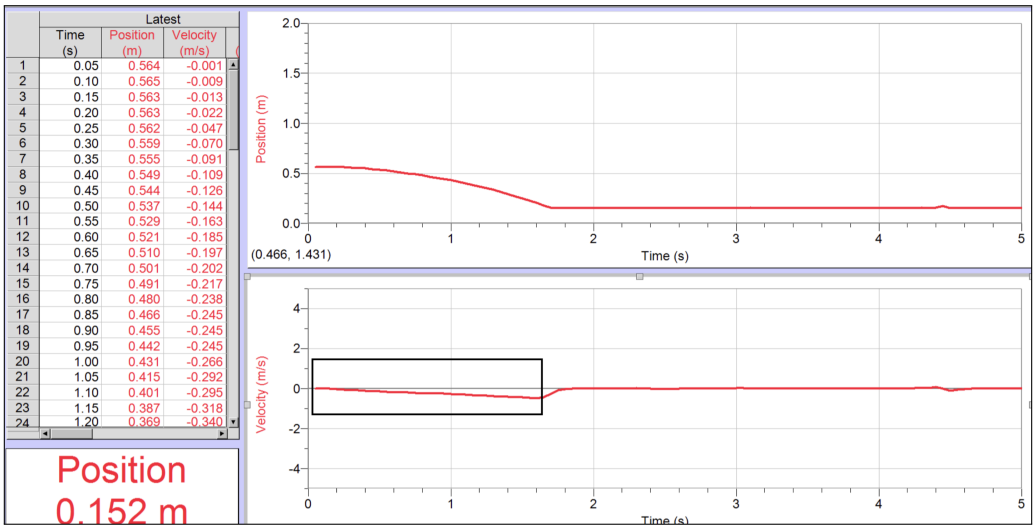


Figure 4: The data-logging software, with the boxed region indicating the section of the velocity-time graph selected to calculate the average acceleration.

12. Repeat steps 10–11 at least 3 times for the same mass ratio M to limit random error and obtain more precise results. There are bound to be human errors, such as applying a non-negligible force to the system when releasing it. It is often difficult to tell whether you have applied such a force, since even slight movements can affect the system; thus, it is difficult to know which trials to disregard. Repeating the experiment multiple times allows us to average out these errors.

13. Repeat steps 6–12 for at least 5 different values of M in the range $0.06 < M < 0.12$. This will allow us to plot a graph of a against M and determine the relationship between the two variables.

4 Data Processing

We now process the raw data collected in Section 3 [p. 4]. The uncertainty in the acceleration is $\pm 0.0001\,\mathrm{m\,s^{-2}}$, corresponding to the precision of the motion sensor. The uncertainty in the mass of the magnetic masses is $\pm 1\,\mathrm{g}$, corresponding to the precision of the scale used.

4.1 Raw Data

Table 1: Raw acceleration data for each trial, ordered by increasing mass ratio M.

m_1 / (g ± 1)	m_2 / (g ± 1)	Trial	a / ($\mathrm{m\,s^{-2}}$ ± 0.0001)
357	408	1	0.0423
		2	0.0595
		3	0.0327
306	357	1	0.1332
		2	0.1262
		3	0.1315
255	306	1	0.2955
		2	0.2993
		3	0.2881
459	561	1	0.4087
		2	0.4144
		3	0.4159
204	255	1	0.4436
		2	0.4741
		3	0.4672

4.2 Processing the M values

Recall the definition of M from (6) [p. 2]:

$$M = \frac{m_2 - m_1}{m_1 + m_2}. \tag{9}$$

We will now calculate the mass ratio M for each row of Table 1 using (9).

An example calculation for the first row of Table 1 [p. 10] is as follows:

$$M = \frac{408 - 357}{408 + 357} \approx 0.0667.$$

We now propagate the uncertainties of m_1 and m_2 to obtain the uncertainty in M. We will do this using half the range of possible values of M,

$$\Delta M = \frac{M_{\max} - M_{\min}}{2}, \tag{10}$$

where $M_{\max}$ and $M_{\min}$ are the largest and smallest values of M allowed by the uncertainties in m_1 and m_2.

Recall the definition of M from (9) [p. 10]:

$$M = \frac{m_2 - m_1}{m_1 + m_2}. \tag{9}$$

Given that M is a function of m_1 and m_2, we can use derivatives to prove two properties of M:

1. M is an increasing function of m_2.
2. M is a decreasing function of m_1.

Together, these imply that the maximum value of M occurs when m_2 is at its maximum and m_1 is at its minimum, and the minimum value of M occurs when m_2 is at its minimum and m_1 is at its maximum. Thus, we find the maximum and minimum values of M by using the extremes of the uncertainties.

Differentiating (9) with respect to m_2 gives

$$\frac{dM}{dm_2} = \frac{(m_2 + m_1)(1) - (m_2 - m_1)(1)}{(m_1 + m_2)^2} = \frac{2m_1}{(m_1 + m_2)^2} > 0,$$

which proves that M is an increasing function of m_2. Furthermore,

$$\frac{dM}{dm_1} = \frac{(-1)(m_1 + m_2) - (m_2 - m_1)(1)}{(m_1 + m_2)^2} = \frac{-2m_2}{(m_1 + m_2)^2} < 0,$$

which proves that M is a decreasing function of m_1.

Thus, to find the maximum value of M, we substitute the maximum value of m_2 (i.e. $m_2 + 1\,\mathrm{g}$) and the minimum value of m_1 (i.e. $m_1 - 1\,\mathrm{g}$) into (9). To find the minimum

value of M, we substitute the minimum value of m_2 (i.e. $m_2 - 1\,\mathrm{g}$) and the maximum value of m_1 (i.e. $m_1 + 1\,\mathrm{g}$) into (9) [p. 11].

Let us now calculate the uncertainty in M for the first row of Table 1 [p. 10]. Given that $m_1 = (357 \pm 1)\,\mathrm{g}$ and $m_2 = (408 \pm 1)\,\mathrm{g}$, we have

$$M_{\max} = \frac{(408+1)-(357-1)}{(408+1)+(357-1)} = \frac{53}{765} \approx 0.0693,$$
$$M_{\min} = \frac{(408-1)-(357+1)}{(408-1)+(357+1)} = \frac{49}{765} \approx 0.0641.$$

Hence, by (10) [p. 11], $\Delta M \approx (0.0693 - 0.0641)/2 \approx 0.003$.

The results for every row are summarised in Table 2.

Table 2: Mass ratio M and its absolute uncertainty ΔM.

M	ΔM
0.067	0.003
0.077	0.003
0.091	0.003
0.100	0.002
0.111	0.004

4.3 Processing the acceleration data

We now calculate the average acceleration ($\bar{a}$) for each mass ratio M from the raw acceleration data in Table 1 [p. 10].

Here is an example calculation for the first mass ratio $M = 0.067$:

$$\bar{a} = \frac{0.0423 + 0.0595 + 0.0327}{3} \approx 0.0448\,\mathrm{m\,s^{-2}}.$$

To find the uncertainty in the average acceleration, we use the maximum and minimum values of the acceleration for each mass ratio M. So, from the first row of Table 1 [p. 10],

$$\Delta\bar{a} = \frac{a_{\max} - a_{\min}}{2} = \frac{0.0595 - 0.0327}{2}\,\mathrm{m\,s^{-2}} = 0.0134\,\mathrm{m\,s^{-2}}.$$

The results for every mass ratio are summarised in Table 3.

Table 3: Average acceleration for each mass ratio M.

M	$\bar{a}$ / $\mathrm{m\,s^{-2}}$
0.067	0.045 ± 0.013
0.077	0.1303 ± 0.0035
0.091	0.2943 ± 0.0056
0.100	0.4130 ± 0.0036
0.111	0.462 ± 0.015

4.4 Processed Data

Combining Table 2 [p. 12] and Table 3 gives the processed data in Table 4.

Table 4: Processed data for the mass ratio M and average acceleration $\bar{a}$.

M	ΔM	$\bar{a}$ / $\mathrm{m\,s^{-2}}$	$\Delta\bar{a}$ / $\mathrm{m\,s^{-2}}$
0.067	0.003	0.045	0.013
0.077	0.003	0.1303	0.0035
0.091	0.003	0.2943	0.0056
0.100	0.002	0.4130	0.0036
0.111	0.004	0.462	0.015

5 Analysis of Results

5.1 Graphical analysis

The processed data is plotted in Figure 5, with M on the x-axis and $\bar{a}$ on the y-axis.

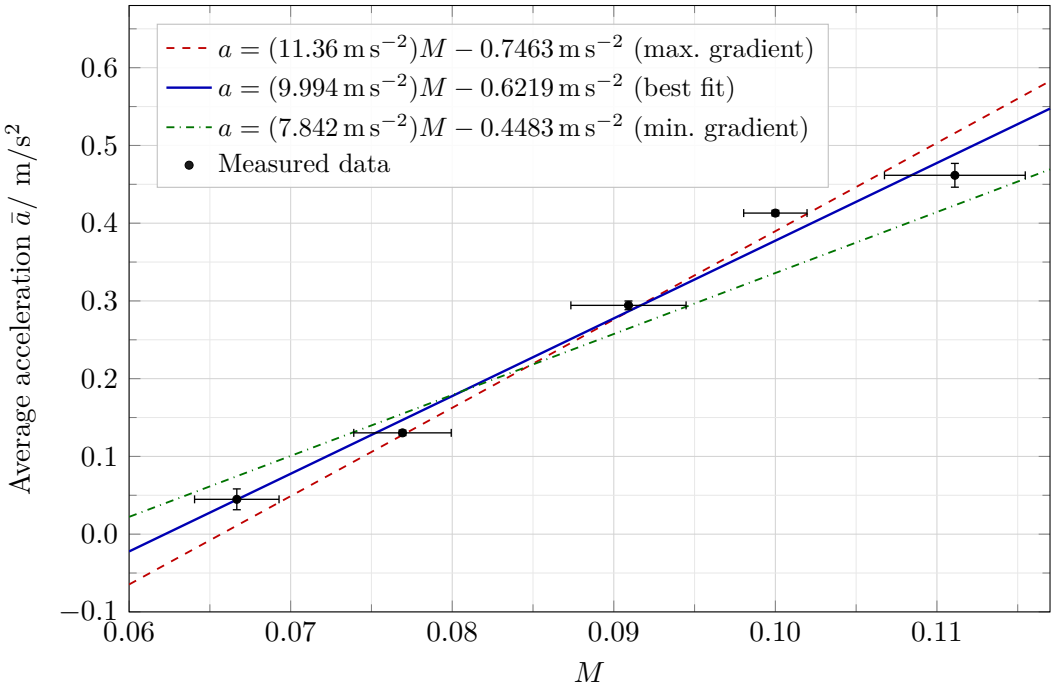


Figure 5: Acceleration against M with error bars.

Note: since $M = (m_2 - m_1)/(m_2 + m_1)$ is a ratio of masses, it is dimensionless. The gradient $\Delta\bar{a}/\Delta M$ therefore has the same unit as $\bar{a}$, namely $\mathrm{m\,s^{-2}}$.

5.2 Slope of the best-fit line

The slope of the best-fit line is $k \approx 9.994\,\mathrm{m\,s^{-2}}$. The maximum and minimum gradient lines in Figure 5 were drawn by eye as the steepest and shallowest straight lines consistent with the error bars of the data points.

We can find the uncertainty in the slope of the best-fit line using the maximum and minimum slopes in Figure 5,

$$\Delta k = \frac{k_{\max} - k_{\min}}{2} \approx \frac{11.36 - 7.842}{2}\,\mathrm{m\,s^{-2}} \approx 1.759\,\mathrm{m\,s^{-2}}.$$

Rounding, we obtain the slope of the best-fit line:

$$k \approx (10.0 \pm 1.8)\,\mathrm{m\,s^{-2}}.$$

The percentage uncertainty in the slope is therefore

$$\frac{\Delta k}{k} \times 100\% = \frac{1.8}{10.0} \times 100\% = 18\%.$$

Recall the equation of the system from (7) [p. 2]:

$$a = gM. \tag{7}$$

Thus, the slope of the best-fit line should theoretically equal $g \approx 9.81\,\mathrm{m\,s^{-2}}$. Our result is consistent with the theoretical prediction, as g is within the uncertainty of our measured slope.

5.3 Intercept of the best-fit line

An unexpected result is that the best-fit line does not pass through the origin, as predicted by theory. This is likely due to friction in the pulley. Figure 5 [p. 14] suggests that there is a minimum M value required to overcome the friction in the pulley.

The equation of the best-fit line in Figure 5 [p. 14] is

$$a = (9.994\,\mathrm{m\,s^{-2}})M - 0.6219\,\mathrm{m\,s^{-2}}.$$

To find the M-intercept, M_{thres}, we set $a = 0$,

$$0 = (9.994\,\mathrm{m\,s^{-2}})M_{\text{thres}} - 0.6219\,\mathrm{m\,s^{-2}}.$$

Solving for M_{thres},

$$M_{\text{thres}} = \frac{0.6219\,\mathrm{m\,s^{-2}}}{9.994\,\mathrm{m\,s^{-2}}} \approx 0.06223.$$

Thus, the M-intercept of the best-fit line is approximately 0.06223. We can also find the uncertainty in the M-intercept using the maximum- and minimum-slope lines in Figure 5 [p. 14].

Setting $a = 0$ in the maximum-slope line, $a = (11.36\,\mathrm{m\,s^{-2}})M - 0.7463\,\mathrm{m\,s^{-2}}$, we have

$$M_{\text{thres,max}} = \frac{0.7463\,\mathrm{m\,s^{-2}}}{11.36\,\mathrm{m\,s^{-2}}} \approx 0.06570.$$

Similarly, setting $a = 0$ in the minimum-slope line, $a = (7.842\,\mathrm{m\,s^{-2}})M - 0.4483\,\mathrm{m\,s^{-2}}$, we have

$$M_{\text{thres,min}} = \frac{0.4483\,\mathrm{m\,s^{-2}}}{7.842\,\mathrm{m\,s^{-2}}} \approx 0.05717.$$

Thus, the uncertainty in the M-intercept is

$$\Delta M_{\text{thres}} = \frac{0.06570 - 0.05717}{2} \approx 0.0043.$$

Hence,

$$M_{\text{thres}} \approx 0.0622 \pm 0.0043.$$

Note that this is very close to the minimum M value we recommended in Section 2 [p. 3] (and experimentally found) to ensure the acceleration of the system is not too high or too low.

6 Conclusion

Our research question was: **How does the acceleration of an Atwood machine with a plastic pulley and masses m_1 and m_2 vary with the mass ratio $M = (m_2 - m_1)/(m_2 + m_1)$, where $m_2 > m_1$ and $0.06 < M < 0.12$?**

We found that the acceleration of an Atwood machine is linearly (but not proportionally) related to the mass ratio $M = (m_2 - m_1)/(m_2 + m_1)$, where $m_2 > m_1$ and $0.06 < M < 0.12$, contradicting our earlier hypothesis of a proportional relationship between the two variables. The slope of the best-fit line was $k \approx (10.0 \pm 1.8)\,\mathrm{m\,s^{-2}}$, consistent with the accepted value of the standard acceleration of gravity, $g = 9.806\,65\,\mathrm{m\,s^{-2}}$ [1], which we take as $g \approx 9.81\,\mathrm{m\,s^{-2}}$. The measured slope differs from the accepted value by $(10.0 - 9.81)/9.81 \approx 1.9\%$, well within its 18% uncertainty. Regardless, our result has a significant 18% uncertainty in the slope. In the following section, we will discuss potential sources of error that may have contributed to this uncertainty. Nevertheless, the literature value of g is within the uncertainty of our measured slope, so our results are consistent with the theoretical prediction. We also found that the M-intercept of the best-fit line was $M_{\text{thres}} \approx 0.0622 \pm 0.0043$, contradicting the theoretical prediction that the best-fit line should pass through the origin. We mainly attribute this to friction in the pulley, which

we ignored when constructing our model. An exact literature value does not exist for this M-intercept, as the minimum M required to overcome friction in the pulley depends on the specific pulley.

Regardless, the non-zero M-intercept and slope have helped answer the research question, suggesting that the acceleration of an Atwood machine is linearly (but not proportionally) related to the mass ratio $M = (m_2 - m_1)/(m_2 + m_1)$, where $m_2 > m_1$ and $0.06 < M < 0.12$. This is commonly accepted, as real Atwood machines have effects arising from friction [2].

7 Evaluation

There is a clear outlier data point at $M = 0.1$. Our data points, accounting for uncertainty, all lie between the maximum and minimum lines in Figure 5 [p. 14], except for this outlier. We investigated this and found that $M = 0.1$ used $m_1 = 459\,\text{g}$ and $m_2 = 561\,\text{g}$, the largest combination of masses used in our experiment (1020 g in total). Furthermore, this was the only data point where the mass difference was 102 g; all other data points used a difference of 51 g. This was also the last datapoint collected. The high total mass and mass difference should not matter mathematically, as the model predicts that only the M value affects acceleration. We therefore believe that the string may have been damaged during the trials. Alternatively, the string may have been initially undamaged, but the final trial's large total mass and mass difference may have caused some of its fibres to tear, causing the string to stretch more than expected and increasing the measured acceleration. To reduce this error, we recommend using a sturdier string, ensuring that it does not stretch or tear and remains closer to the idealised model. The string should be replaced after each trial; however, the material of the string must be held constant to ensure that the coefficient of friction remains constant. This error could significantly affect the results. If the string is damaged during the first trial and all successive trials use this faulty string, the overall results may become less accurate. In our experiment, however, this effect likely only impacted the final trial; thus, the overall impact was likely minor.

Another source of error is the uncertainty of the acceleration of the system. Recall the uncertainty values from Table 4 [p. 13], reproduced in Table 5 [p. 18]:

Table 5: Processed data for the mass ratio M and average acceleration $\bar{a}$.

$M \pm \Delta M$	$\bar{a}$ / $\mathrm{m\,s^{-2}}$	$\Delta\bar{a}$ / $\mathrm{m\,s^{-2}}$
0.067 ± 0.003	0.045	0.013
0.077 ± 0.003	0.1303	0.0035
0.091 ± 0.003	0.2943	0.0056
0.100 ± 0.002	0.4130	0.0036
0.111 ± 0.004	0.462	0.015

We can write the uncertainties as percentages. For example, for the first data point, the percentage uncertainty is

$$\frac{\Delta\bar{a}}{\bar{a}} = \frac{0.013}{0.045} \approx 29\%.$$

These are summarised in Table 6.

Table 6: Percentage uncertainties in the average acceleration $\bar{a}$.

$M \pm \Delta M$	$\Delta\bar{a}/\bar{a}$
0.067 ± 0.003	29%
0.077 ± 0.003	2.7%
0.091 ± 0.003	1.9%
0.100 ± 0.002	0.87%
0.111 ± 0.004	3.2%

For the first data point, the uncertainty in the average acceleration is 29%, which is very high. The uncertainty starts decreasing until it reaches a minimum (0.87%) at the second last data point, and then increases again to 3.2% for the last data point. For the first data point, the likely reason was the let-go mechanism. We attempted to hold the mass at rest and we then let it go. However, because the acceleration of the system was very low, it is likely that we applied a non-negligible force to the system when letting it go, which would have drastically affected the measured acceleration of the system. Such a force matters less for the other data points because the acceleration of the system is higher, meaning that the force we apply to the system when letting it go is negligible compared to the force of gravity acting on the system. For the final data point, when M is sufficiently high, it is possible that the system accelerated down so quickly that the motion sensor did not have enough time to take a sufficient number of measurements to accurately measure the acceleration of the system, which would have increased the uncertainty in

the measured acceleration. However, this may be a rather unrealistic argument as the uncertainty for the final data point is only 3.2%, which is not too high. To fix this, we suggest using magnetic weights with finer graduations. We only had access to magnetic weights of 51 g and 102 g, which limited the number of combinations of masses we could use. Technically more would have been possible but those combinations would require very high total masses which would have likely led to string damage. Thus, more combinations of magnetic weights (e.g. 10 g, 20 g, 30 g, 40 g) would have allowed us to test more M values in the range $0.06 < M < 0.12$ without having to use very high total masses, which would have reduced the uncertainty in the measured acceleration because we would not need to be very close to $M \approx 0.06$ or $M \approx 0.12$ to get a sufficient number of data points in the range $0.06 < M < 0.12$. The uncertainty in the measured accelerations is likely minimal as the uncertainties are already fairly small. However, the suggested fix could also enable us to gain more data points in the range $0.06 < M < 0.12$, which would allow us to more accurately determine the relationship between M and a.

In terms of the errors in the experiment, friction likely had the largest relative impact. The effect of friction is visible in our results, as the best-fit line does not pass through the origin. Instead, there is a minimum M value required to overcome the friction in the pulley. To account for this limitation, we suggest extending the mathematical model to include friction. A more accurate model may enable us to gain further insight through linearisation; it may be possible, for example, to derive the dynamic or static coefficient of friction and compare it against literature values. The reason this is our suggested fix is that attempting to reduce friction (e.g., with lubrication) will likely still not eliminate it; thus, it is more practical to account for it in the mathematical model, especially as it may enable us to derive further insight into the system.

It may be noted that the pulley's moment of inertia likely had some effect on the system; however, the mathematics of a pulley's moment of inertia goes beyond the IB syllabus, so it is not considered here.

The relative impact of the outlier data point is not as significant but has some effect. The best-fit line ended up with a higher slope to accommodate this data point (because the acceleration was unexpectedly high during this trial), which is likely why our result for g [$(10.0 \pm 1.8)\,\mathrm{m\,s^{-2}}$] was larger than the accepted value of $g \approx 9.81\,\mathrm{m\,s^{-2}}$. Finally, the relative impact of the uncertainty in the measured accelerations is not as significant, as the uncertainties are very small (except for the first data point).

References

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